

Fermi Balls:

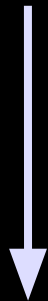
How to make a
Primordial Black Hole

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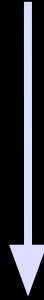
01

PBH Dark Matter

DARK MATTER?

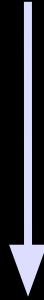


DARK MATTER?



1. DARK

DARK MATTER?

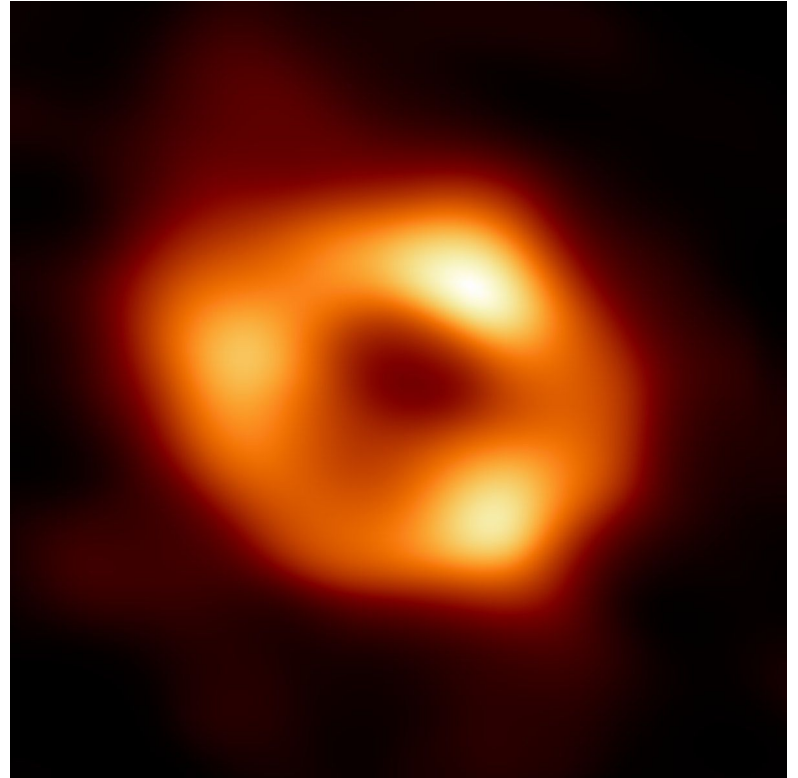


1. DARK

2. MATTER

BLACK HOLES

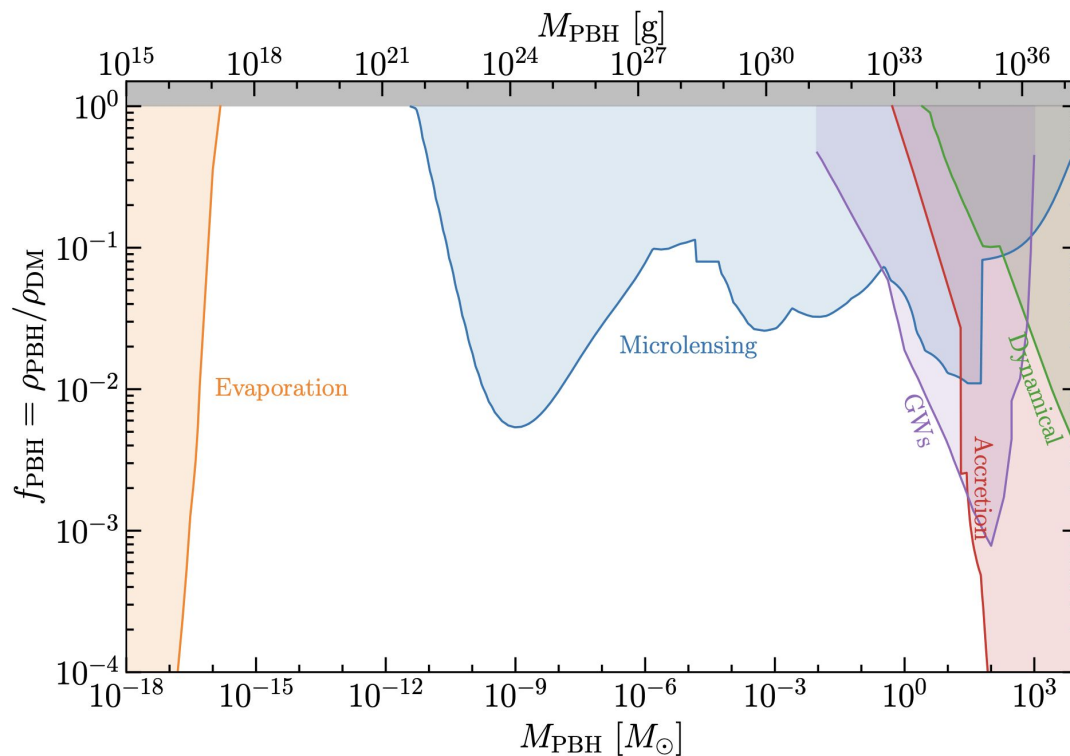
Supermassive	• 4 billion $M_{\odot}$ • Too big
Stellar	• Over 3 $M_{\odot}$ • Abundance too low
Primordial	• Zel'dovich and Novikov (1967) • Hawking (1971)



Sgr A*, Event Horizon Telescope, 2022

FORMATION

- Horizon grows after inflation, letting in density perturbations
- Perturbations can collapse gravitationally, forming black holes
- Wide range of possible masses – roughly the horizon mass at the time they were formed



Green, 2024

02

Fermi Ball Formation

A SCALAR FIELD

A spin zero boson ϕ , coupled to a Dirac fermion χ

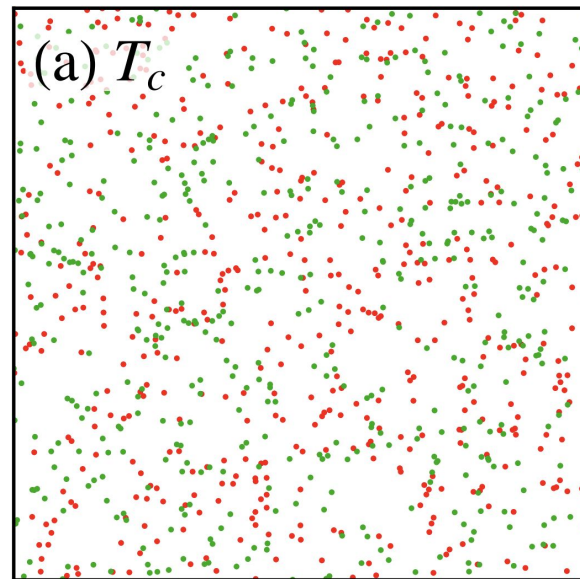
LAGRANGIAN

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - U(\phi) + \bar{\chi}i\not{\partial}\chi - g_\chi\phi\bar{\chi}\chi$$

DEGENERATE VACUA

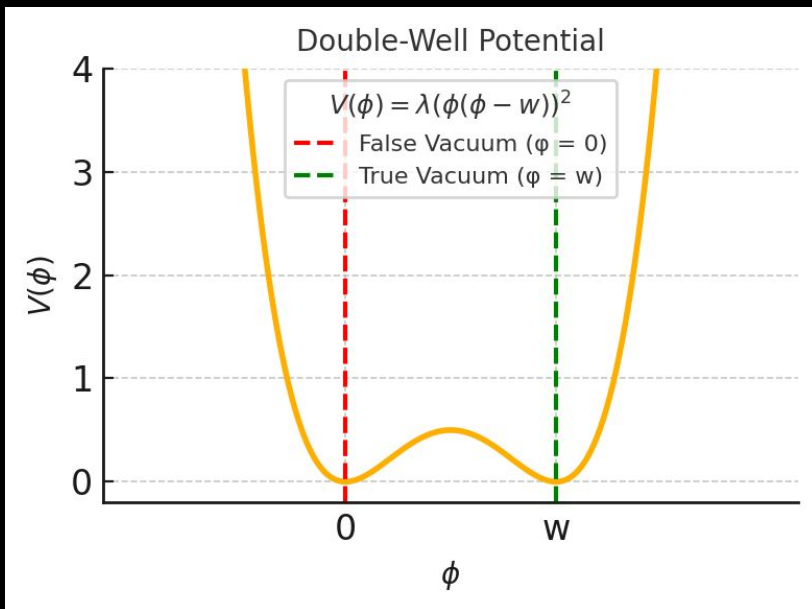
- Symmetric potential
- Two degenerate vacuum states:
 $\langle\phi\rangle=0$ and $\langle\phi\rangle=w$
- Universe stays in the “false vacuum” $\langle\phi\rangle=0$

TEMPERATURE: T_c

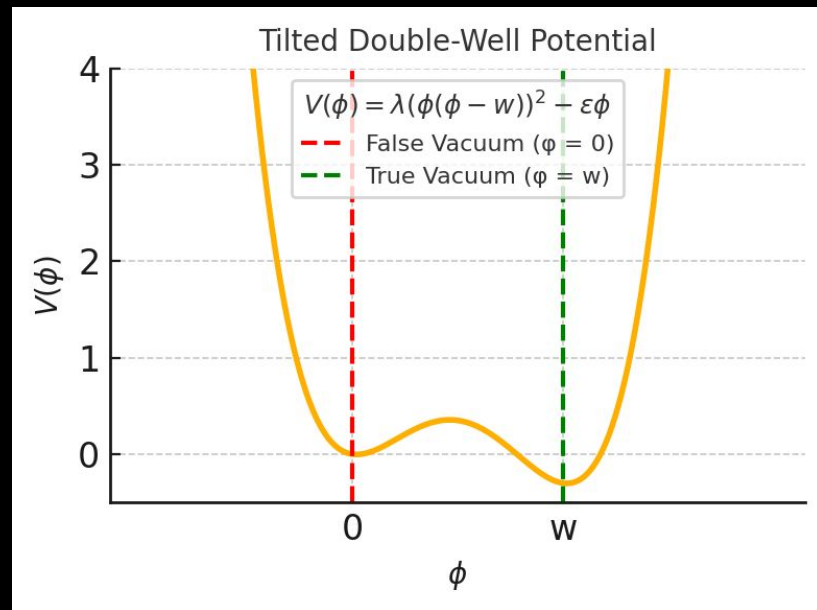
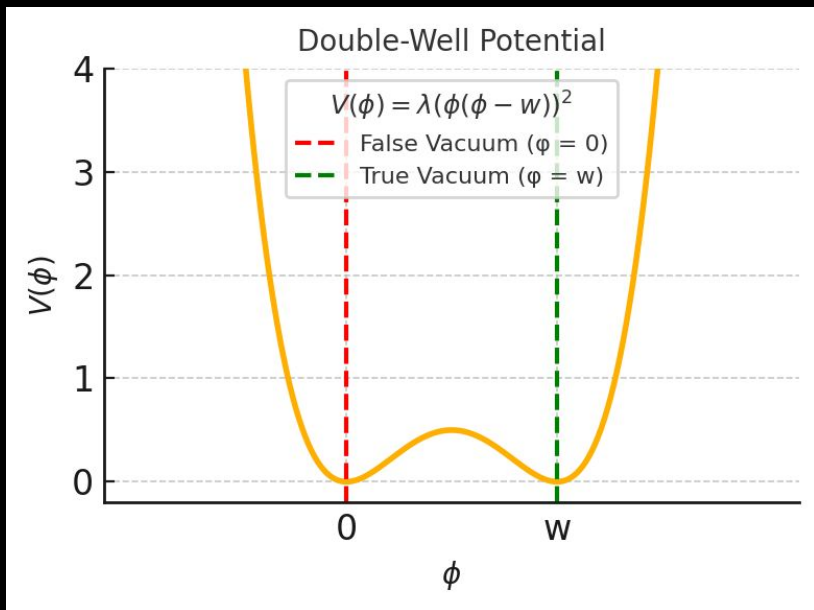


Kawana and Xie, 2021

$$U(\phi) = U(\phi, \mathbf{T})$$



$$U(\phi) = U(\phi, \mathbf{T})$$



A FIRST ORDER PHASE TRANSITION

True vacuum bubbles form within the false vacuum

BUBBLE NUCLEATION

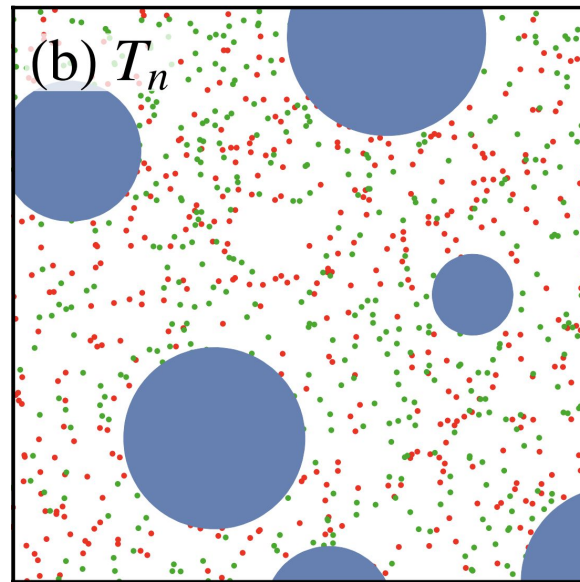
- ϕ can tunnel from $\langle\phi\rangle=0$ to $\langle\phi\rangle=w$
- Probability of tunneling: $\Gamma \sim T^4$
- Nucleation happens at T_n when

$$\frac{\Gamma(T_n)}{H^4(T_n)} \approx 1$$

FERMIONS BECOME TRAPPED

- Fermion mass = $g_\square \phi$
- Massless in false vacuum, mass $g_\square w$ in the true vacuum
- Cannot penetrate into true vacuum

TEMPERATURE: T_n

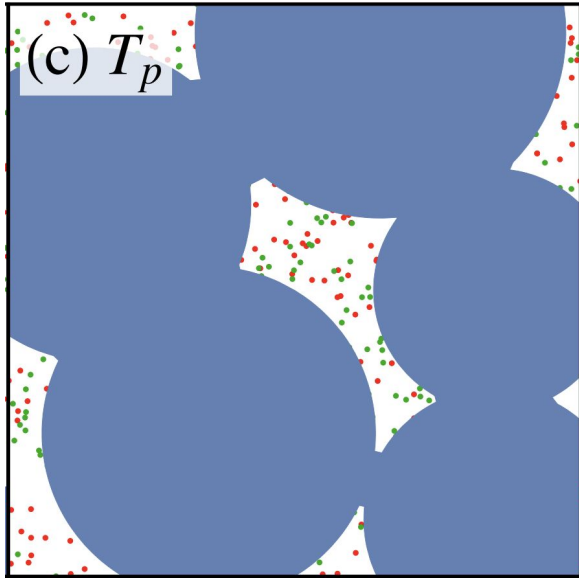


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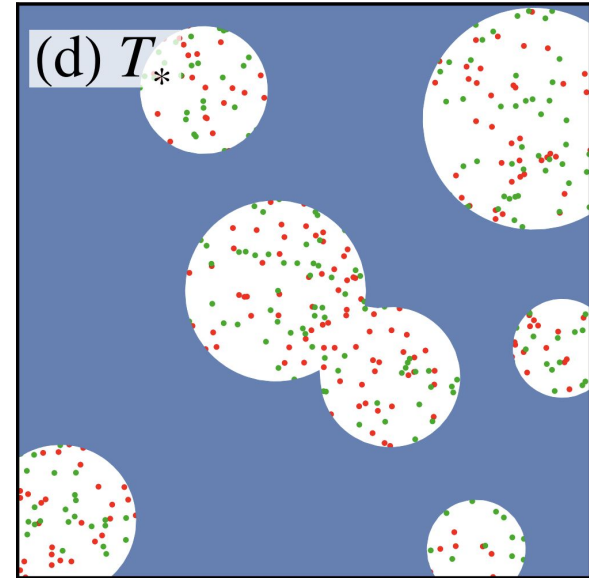
BUBBLE PERCOLATION

True vacuum bubbles form an “infinite connected cluster”

TEMPERATURE: T_p



TEMPERATURE: T_*



Critical Remnants

FERMI BALLS FORM

Fermions compress into compact objects

STABILITY

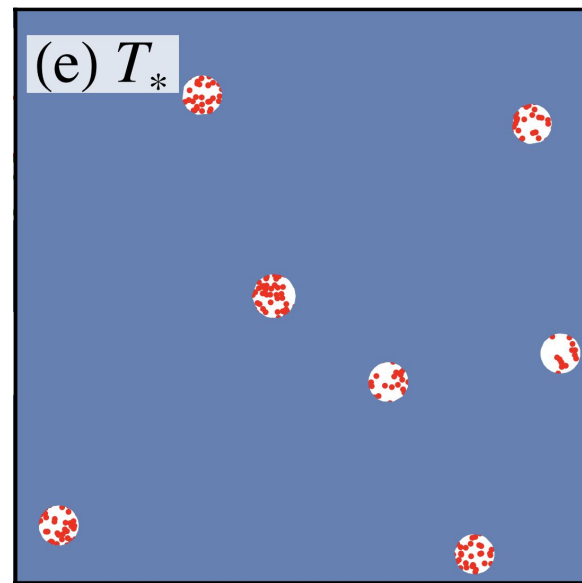
- $\bar{\psi}\psi$ annihilate – asymmetry
- Shrinking stops when vacuum pressure $U_0(T_*)$ balances degeneracy pressure

$$U_0(T) \equiv U(0, T) - U(w(T), T)$$

FERMI BALL PROFILE

$$E_{FB} = \frac{3\pi}{4} \left(\frac{3}{2\pi} \right)^{\frac{2}{3}} \frac{Q_{FB}^{\frac{4}{3}}}{R} + \frac{4\pi}{3} U_0(T_*) R^3$$

TEMPERATURE: T_*



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03

Collapse into PBHs

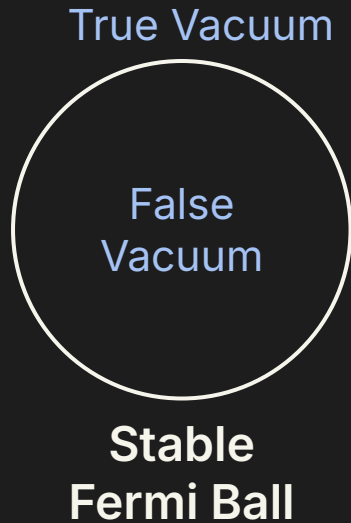
The Yukawa Interaction

How do the Fermions talk to each other?

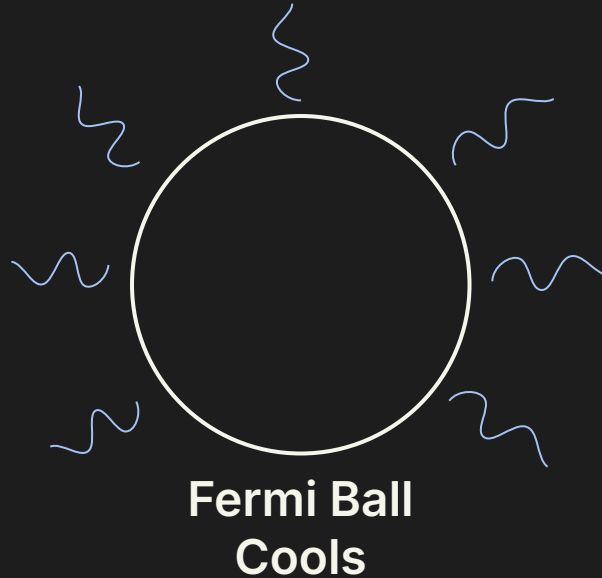
$$V(r) = -\frac{g_\chi^2}{4\pi r} e^{-M_\phi r} \quad M_\phi^2 = \left. \frac{d^2 U(\phi, T)}{d\phi^2} \right|_{\phi=0} = \mu^2 + cT^2$$

As the universe cools, $L_\phi = M_\phi^{-1}$ increases, and the Yukawa potential becomes significant.

Fermi Ball Collapse



False vacuum inside,
Yukawa potential
insignificant



$\square \rightarrow \square \bar{\bar{\nu}}$, reaction
rate greater than H ,
tracks temperature
of universe



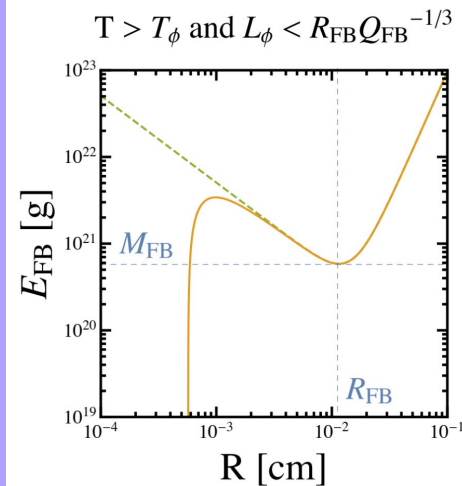
As its range
increases, it
dominates over
degeneracy
pressure

PBHs

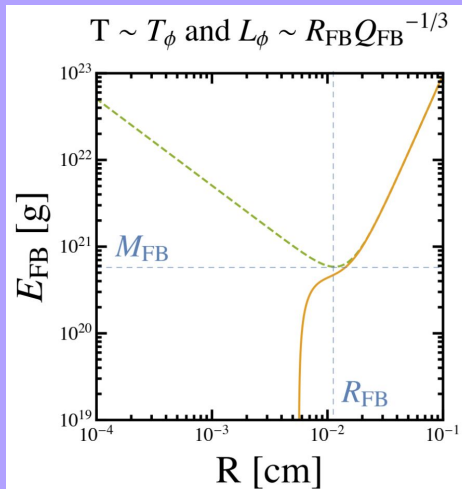
$$E_{FB}^{Yukawa} \approx -\frac{3g_\chi^2}{20\pi} \frac{Q_{FB}^2}{R} \times \frac{5}{2} \left(\frac{L_\phi}{R} \right)^2$$

$$E_{FB}^{Fermi} = \frac{3\pi}{4} \left(\frac{3}{2\pi} \right)^{2/3} \frac{Q_{FB}^{4/3}}{R}$$

When $L_\phi/R_{FB} \sim Q_{FB}^{-1/3}$, Yukawa energy dominates over degeneracy pressure, and Fermi balls collapse to black holes.



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SUMMARY

01 A SCALAR FIELD

False vacuum
Dirac fermions

02 A PHASE TRANSITION

Tunnels to true vacuum
Fermions stay in false
vacuum

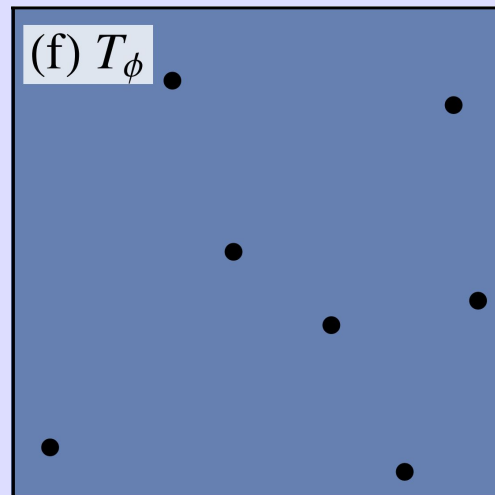
03 STABLE FERMI BALLS

Vacuum pressure
balances degeneracy
pressure

04 FERMI BALLS COLLAPSE

Yukawa potential
dominates degeneracy
pressure
Collapses to black hole

TEMPERATURE: T_ϕ



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Questions

References

Green, A. M. (2024). *Primordial black holes as a dark matter candidate—A brief overview*

(arXiv:2402.15211). arXiv.

Hong, J.-P., Jung, S., & Xie, K.-P. (2020). *Fermi-ball dark matter from a first-order phase transition*

(arXiv).

Kawana, K., & Xie, K.-P. (2021). *Primordial black holes from a cosmic phase transition: The*

collapse of Fermi-balls (arXiv:2106.00111). arXiv.

Lu, Y., Picker, Z. S. C., Profumo, S., & Kusenko, A. (2025). *Black Holes from Fermi Ball Collapse*

(arXiv:2411.17074). arXiv.

Macpherson, A. L., & Campbell, B. A. (1995). *Biased discrete symmetry breaking and Fermi balls.*

Macpherson, A. L., & Pinfold, J. L. (1994). *Fermi Ball Detection* (arXiv:9412264). arXiv.

Ogure, K., Yoshida, T., & Arafune, J. (2002). *Stable Neutral Fermi Ball* (0212332). arXiv.